

OpenMDBench: Layering the LLM Only Where It Pays

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Abstract

Hybrid stacks that place an LLM deliberator above a small-model executor are the convergent engineering answer for decision problems that separate a slow, semantic deliberation layer from a fast, continuous execution layer; multi-domain adversarial tasks are a primary instance. Yet *when* this layering pays, and *why*, still lacks benchmark-grade evidence. We formalize the two-tier stack as a hierarchical semi-MDP and derive a *conditional layering framework* with three measurable conditions: (i) core difficulty in LLM-strong decision forms, (ii) a goal interface carrying information the executor cannot reconstruct, and (iii) monolithic baseline headroom. **OpenMDBench**, a distributed open benchmark with central simulation and local inference, validates the framework on two instantiations sharing one interface stack. On a semantically enhanced grid, where decision-form match fails and baseline headroom is insufficient, goal-dose ablations and a live-LLM counterfactual show the interface has a causal contribution yet deployed hybrids trail the strongest pure baseline, the framework’s predicted boundary. On a gate-admitted 14-scenario high-fidelity suite, admitted by the pre-registered design-side gates, layered stacks beat the Rule baseline in 11 and the RL baseline in 12 of 14 scenarios, while end-to-end LLM control trails decisively. A mechanism probe shows layered value does *not* require an LLM information advantage: the LLM discriminator underperforms a rule baseline, yet layering still wins; the evidence is more consistent with semantic goal decomposition than with an LLM information advantage. Replay-gated counterfactual attribution localizes failures to planning, execution, or the interface. Our results support a conditional placement principle: in the evaluated cross-domain setting, LLMs are most effective as deliberators above mature executors when decision-form match, interface carrying, and baseline headroom jointly provide room for layered gains.

Keywords

multi-agent systems, evaluation benchmark, large language models, hybrid LLM–small model architecture, distributed open evaluation, multi-domain heterogeneous systems

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1 Introduction

Two problems, two decision layers. Decision problems of interest separate a *deliberation* problem—whom to watch, what to intercept, how to respect interaction protocols, at second-to-minute cycles over semantic, open-ended states—from an *execution* problem—control, tracking, sensing, reactive avoidance, at millisecond-to-second cycles over continuous spaces. In our multi-domain coastal-protection instantiations, the layer boundary coincides with physical-domain boundaries (aerial, surface, shore, underwater). Pure LLM agents cannot meet execution latency and cost budgets; pure RL agents cannot ingest open-ended language objectives or adversarial deception. **Hybrid stacks—LLMs as deliberators, small models as executors—are the community’s convergent response** [3, 11, 14, 46].

Three questions, no ground to answer them. If hybridization is the future, benchmark-grade evidence for three questions is conspicuously absent: **(Q1) Payoff**—does a hybrid outperform the best pure paradigm, and where does it *underperform*? **(Q2) Necessity**—under what measurable task properties is layering *structurally necessary*? **(Q3) Bottleneck**—when a hybrid fails, is the fault in planning, execution, or the interface?

The obstacle is infrastructural: the dominant MARL-era benchmarks—from SMAC [27] to LLMarena [5]—inherit the centrally hosted evaluation model. For hybrid systems this fails computationally (hosting billion-parameter LLMs centrally is prohibitive), institutionally (proprietary weights cannot be uploaded), and architecturally (predefined interfaces constrain the composition space hybrid research explores)—so architectures are built with no shared ground to compare them and no instrumentation to tell *why* one wins.

OpenMDBench. We resolve the paradox with a requirements-driven pipeline—formalize, build, instrument, validate—whose outputs map one-to-one onto three contributions. Validation runs on two instantiations sharing one interface stack: a semantically enhanced grid maps the law’s boundary conditions over ~1,000 episodes of a complete planner×executor factorial, and a high-fidelity adversarial suite compares five stacks on 14 deception-enabled scenarios under a unified no-intelligence regime. All numbers derive from released per-episode traces.

Contributions. Our contributions are threefold, mapping one-to-one onto the pipeline and answering Q1–Q3 in order: the framework predicts *when* layering can pay (Q2), the toolbox localizes *why* a given stack falls short (Q3), and the platform supplies the ground on which both claims are measured (Q1).

(C1) The conditional layering framework, validated on both sides (Section 3). From an HSemi-MDP formalization of the two-tier stack and its coupling structure, we derive a *conditional layering law* with three measurable conditions—decision-form match, interface carrying, baseline headroom—and three falsifiable predictions (Figure 1). The grid *confirms the boundary prediction*:

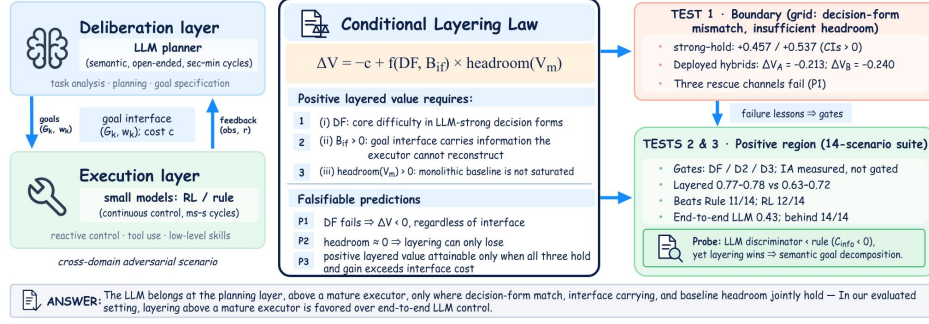


Figure 1: The conditional layering law and its three tests. The two-tier decision stack (left) feeds the law’s three measurable conditions (center): decision-form match (DF), interface carrying (B_{if}), and baseline headroom. Test 1 (top right): on the grid boundary regime where decision-form match fails and headroom is insufficient, the goal interface is causally necessary (strong-hold: $+0.457 / +0.537$, CIs > 0) yet deployed hybrids trail the strongest pure baseline ($\Delta V_A = -0.213$; $\Delta V_B = -0.240$); three rescue channels all fail (P1), and the failure lessons harden into pre-registered gates DF/D2/D3. Tests 2 and 3 (bottom right): on the gate-admitted 14-scenario high-fidelity suite admitted by the pre-registered design-side gates, layered stacks achieve 0.77–0.78 vs. 0.63–0.72 for the monolithic baselines, beating Rule in 11 and RL in 12 of 14 scenarios; end-to-end LLM control trails at 0.43, behind on 14/14 scenarios. A mechanism probe shows the LLM discriminator underperforms the rule baseline ($C_{info} < 0$), yet layering still wins—the evidence is more consistent with semantic goal decomposition than with an LLM information advantage.

where decision-form match fails and headroom is insufficient, deployed hybrids trail the strongest pure baseline even though the interface remains causally useful. The gate-admitted high-fidelity suite provides evidence for the positive-region hypothesis under the pre-registered design-side admission protocol: admitted by the pre-registered design-side gates, layered stacks beat Rule in 11 and RL in 12 of 14 scenarios. A mechanism probe shows layered value does *not* require an LLM information advantage—locating value in semantic goal decomposition, not perception asymmetry.

(C2) A diagnostic toolbox that turns failure analysis into measurement (Sections 5–6). A replay-gated counterfactual-attribution method decomposing gaps into planning/execution/synergy terms, plus a goal-dose ablation operationalizing B_{if} causally. Validated on anchored faults (20/20) and live LLM streams (10/10); they convert where a hybrid fails into per-episode quantities.

(C3) OpenMDBench, the experimental substrate (Section 4). A distributed open platform (simulation central, inference local via polling API) removing the three frictions excluding hybrid systems; it ships two instantiations sharing one interface stack, a 36-scenario family, layered metrics, and trace release. The platform is the precondition on which C1–C2 are measurable, and together they answer where the LLM should sit.

Paper organization. Section 2 positions related work; Section 3 formalizes problem and coupling theory; Sections 4–5 present platform and diagnostics; Section 6 validates; Sections 7–8 discuss and conclude. Results derive from released per-episode traces and manifests.

2 Related Work

MARL benchmarks. Classic cooperative MARL benchmarks [7, 27] and continuous-control environments such as Google Research

Football [16] standardize comparison in single domains with homogeneous agents; heterogeneity [6, 10, 41], application [20, 30], social-dilemma [17], and standardization [4, 13, 33] remain single-domain and platform-hosted; BenchMARL [4] further advances standardized and reproducible benchmarking across MARL algorithms and environments. Existing work has demonstrated LLM-MARL hierarchical architectures [3, 11, 14, 46], but existing benchmark suites do not systematically evaluate *when* such layering is beneficial, *when* it fails, or *which* layer is responsible for the failure.

LLM agent benchmarks. General LLM-agent benchmarks—AgentBench [18], WebArena [45], GAIA [21]—measure task completion across web, tool-use, and knowledge domains. Interactive-LLM multi-agent benchmarks [5, 12, 19, 40, 47] evaluate LLM agents but run the LLM on the platform side—the inverse of our distributed paradigm—and none spans multiple physical domains or evaluates hybrid stacks; surveys map this fragmented landscape [23]. Position work argues executable agent evaluation suffers an *identification problem*: benchmark scores and leaderboard rankings can be sensitive to seemingly minor evaluation conventions [1], motivating tighter control of non-model evaluation components [1]—the agenda OpenMDBench implements (C3).

Hybrid LLM-small model architectures. Compositional designs—CICERO [2], LLM controllers over specialized models [29], reasoning-action interleaving [42], cognitive-architecture frameworks [31], hierarchical HTN-LLM planners [24], and LLM-RL combinations [3, 11, 14, 46]—converge on the same placement: the LLM deliberates above a mature executor, in line with the LLM-Modulo position that LLMs cannot autonomously plan but can *help* planning [15] and critical audits of end-to-end LLM planning [35]. The hierarchical structure traces to the

Table 1: OpenMDBench vs. representative benchmarks.

Benchmark	Domain	Agent type	Execution	LLM	Hybrid	NL
SMAC [27]	Single	Homog. MARL	Local environment	✗	✗	✗
LLMArena [5]	Single	Pure LLM	Benchmark-side	✓	✗	✓
MultiAgentBench [47]	Diverse	Pure LLM	Benchmark-side	✓	✗	✓
SPIN-Bench [40]	Multi-domain	Pure LLM	Benchmark-side	✓	✗	✓
OpenMDBench	Multi-domain	Hybrid	Central sim / local inference	✓	✓	✓

Options/Semi-MDP framework [32]; we add a semantic goal interface and derive empirical conditions for when the upper layer pays. What this line lacks is benchmark-grade evidence for *when* the placement pays—the gap our conditional layering law fills (C1).

Adversarial and security evaluation. Network-security RL environments such as CyberBattleSim [36] study deception in single-domain settings with RL agents; LLM cyber agents [26] deploy LLMs directly as attackers in NetSecGame, and Defender-Bench [44] evaluates LLM language agents across offense, defense, and knowledge tasks—yet all remain single-domain, treat LLM agency as beneficial by default, and none measures when a layered stack outperforms a monolith. Our asymmetric coastal-protection suite extends adversarial evaluation across physical domains (aerial, surface, shore, underwater) with a standardized scripted opponent and deception-by-construction templates, explicitly testing the conditional value of layering rather than assuming it.

Multi-agent attribution and failure diagnosis. Counterfactual credit assignment originates in MARL with COMA [9]. Subsequent tools decompose outcomes across agents [34], quantify emergent synergy [38], localize Patient-0 failures [28], and score module contributions inside an LLM agent [39]—all explain *agents, policies, or modules*; none is replay-gated, validates on real LLM streams, or ships as a standard protocol. To our knowledge, our $\Delta P/\Delta E/\Delta I$ counterfactual contribution decomposition is the first *replay-gated* protocol localizing planning-vs-execution contributions on *live* LLM streams (10/10) and anchored faults (20/20) as a built-in benchmark component (C2).

Distributed and federated evaluation. Distributed computation in RL has been training-side [8, 22, 25, 37]; OpenFedLLM [43] federates LLM *training*. To our knowledge, no prior benchmark runs simulation centrally while agent inference runs locally—a new evaluation paradigm for hybrid systems.

Comparison. Table 1: to our knowledge, no existing benchmark combines multi-domain heterogeneity, hybrid-architecture support, natural-language task interfaces, and open evaluation.

3 Problem Definition: Decision-Stack Bifurcation and Formalization

3.1 The Two Layers

The problem bifurcates by cycle and state type: a *deliberation layer* over semantic state at second-to-minute cycles (mission understanding, cross-domain allocation, intent inference, protocol compliance, replanning) and an *execution layer* over continuous state at millisecond-to-second cycles (control, tracking, sensing, reactive tactics)—a property of the problem, not the solver.

3.2 HSemi-MDP Formalization

We formalize as a **hierarchical semi-MDP** $M = \langle M_d, \{M_e^k\}, G \rangle$: an upper deliberation Semi-MDP M_d over semantic state (allocation, task progress, opponent belief, rules) with event-driven goal-assignment actions at random inter-decision intervals; a lower execution-MDP family M_e^k per domain with continuous dynamics and rewards parameterized by goals (G_k, w_k) handed down from above. The interface carries explicit cost $c \geq 0$ —goal-conditioned execution is weaker than an unconstrained monolith when goals carry no exploitable information, reconciling why layering can hurt. A scripted opponent standardizes scenarios; its strength knobs calibrate headroom.

3.3 Cross-Domain Coupling and the Layering Law

A key structural parameter is **cross-domain coupling**—how much upper-layer decisions coordinate across domains. We measure it as effective coupling, the mean number of domains touched per allocation decision, observable from execution traces; intrinsic task coupling (minimum required domain count $\times$ inter-domain dependency strength) bounds it from above. Coupling enters the law indirectly: it raises the ceiling on what the interface can carry (a goal stream coordinating more domains lifts the upper bound on B_{if}) and shapes which decision forms dominate planning, modulating DF.

Operationalizing the coupling terms. Two quantities anchor the law. **Interface information** B_{if} , the effective information carried by the goal interface, is operationalized *causally* as the paired performance cost of disabling the interface (the goal-dose ablation of Section 5.2), not by an observational estimate. **Channel information asymmetry** C_{info} , the information asymmetry between the two observation channels, is probed *behaviorally* through discrimination performance on the public channel. C_{info} is *not* a condition of the law but a competing mechanism hypothesis (whether layered value requires an LLM information advantage), tested in Section 6.9 and reframed there as a reported diagnostic rather than a gate. Both are grid-side instruments; the high-fidelity suite relies on the pre-registered gates (Section 6.7).

The layering law. Let V_m be the best *pure-execution* (monolithic) baseline and $\Delta V = V_{hybrid} - V_m$. Under the time-scale separation of Section 3.1, the hybrid value decomposes into an executor floor V_m , an interface constraint cost $-c \leq 0$ (goal-conditioned execution cannot exceed an unconstrained monolith unless goals carry exploitable information), and a planner’s gross gain. Two forces bound that gain: the executor absorbs at most a fraction $f(DF, B_{if})$ of it—zero when the core difficulty is not in LLM-strong decision forms (DF) or the interface carries nothing the executor cannot reconstruct (B_{if})—and the gain cannot exceed the headroom $\text{headroom}(V_m)$ the baseline leaves below the oracle V^* . Assembling the two bounds yields the conditional form

$$\Delta V = -c + f(DF, B_{if}) \text{headroom}(V_m) \quad (1)$$

with $\text{headroom}(V_m) = (V^* - V_m)/(V^* - V_{min})$, the normalized gap to oracle—the worst-baseline denominator removes absolute-scale

dependence, so headroom is comparable across scenarios of different difficulty—and $f \geq 0$ increasing in both arguments. Here DF denotes *decision-form match*: the task’s core difficulty must reside in semantic decision forms the LLM handles well (allocation, intent inference, protocol compliance), not in spatial micro-control where small models are stronger.

Why each condition is necessary. The three conditions are not independent heuristics; each is motivated by the HSemi-MDP structure and the semantic-goal interface design. **(i) Decision-form match.** If the core bottleneck lives in the execution layer (geometric micro-control), the goals handed down by the upper layer are information the executor cannot exploit—it is already saturating on kinematics, not waiting for guidance—so $B_{if} \rightarrow 0$ and $f \rightarrow 0$, leaving $\Delta V = -c$: layering adds only interface cost. **(ii) Interface carrying.** If the goal interface does not carry information the executor could not reconstruct itself—the LLM merely restates state the executor already observes—then the upper layer contributes nothing beyond the monolithic baseline, and again $\Delta V = -c$. A third route to the same outcome is an *executor floor*: when the executor saturates at a performance floor, goals cannot rescue it and $B_{if} \rightarrow 0$ regardless of what the interface carries (Section 6.4). **(iii) Baseline headroom.** If the monolithic baseline is already near oracle (headroom ≈ 0), even a positive $f(DF, B_{if})$ is multiplied by headroom ≈ 0 , and again $\Delta V \approx -c$. The law’s falsifiable content is therefore that $\Delta V > 0$ requires all three to hold jointly (necessary, not sufficient: the planner gain must also exceed the interface cost c); violating any one reduces the second term to zero and layering can only lose.

Eq. 1 fixes the *directional* structure; we deliberately do not fit f or estimate c in absolute terms (Section 7.2). Prior unconditional formulations (“advantage scales positively with coupling”) omit decision-form match and headroom and are refuted by our boundary-regime experiments: the interface is causally necessary (Section 5.2), yet at a near-saturated baseline with a decision-form mismatch both deployed hybrids sit significantly *below* it—layering only loses where the conditions fail and pays where they jointly hold (Sections 6.4–6.8). Figure 1 summarizes the law and the three falsifiable predictions below.

Performance accounting. A 2×2 deployment accounting identity (planner/executor swap contrasts P, E, I vs. the strongest pure baseline) fixes the threshold layering must clear (Table 2; full derivation: App. I).

Design corollaries (pre-registered acceptance gates). The law converts into falsifiable environment requirements enforced before any full-scale run: **(DF)** decision-form match (operationalized as a myopic greedy allocation script scoring $< 90\%$ of the rule baseline), **(D2)** headroom (best pure baseline SR $\in [40\%, 85\%]$), and **(D3)** interface budget (goal-dose effects exceeding seed-level variance). Information asymmetry (IA) is measured but not gated (Section 6.9); Section 6.6 shows the DF-violation case, and Section 6.7 embeds DF/D2/D3 into the high-fidelity protocol.

Three falsifiable predictions. The law converts into three testable predictions: **(P1)** at a decision-form mismatch (DF fails), the planner-side gain vanishes, so $\Delta V \leq -c$; **(P2)** at a saturated baseline (headroom ≈ 0), layering can only lose; **(P3)** positive

layered value is attainable only in the joint-positive region of all three conditions; outside that region, positive value is structurally ruled out under the model assumptions. Section 6 tests each: P1–P2 on the grid’s boundary regime, P3 on the gate-admitted high-fidelity suite.

4 The OpenMDBench Platform

4.1 Distributed Open Evaluation Architecture

The **Central Simulation Server (CSS)** maintains environments, generates briefings, executes actions, and computes metrics. **User-Local Hybrid Agents (ULHAs)** run all inference locally behind a standardized **polling API** (RESTful HTTP, JSON; App. B); simulation advances asynchronously of inference and sessions are resumable. This removes the three frictions excluding hybrid systems from prior benchmarks: no weights cross the trust boundary, users bear inference cost, and the composition space stays open. Integrity is enforced by protocol rather than trust: rate limits and action-pattern checks bound the trust model, manifest-hash freezing fixes versions, and determinism gates plus channel-fairness audits equalize what each stack sees (App. B, App. H).

4.2 Scenario Family and Task Suite

We instantiate an **asymmetric coastal-protection** family: Red (defender, evaluated) fields UAVs, USVs, shore radar, and AUVs (heterogeneous, complementary sensing); Blue (scripted opponent) uses tactic templates with calibrated strength knobs (App. C). 36 scenarios across 5 categories (reconnaissance, tracking & surveillance, interception & response, zone control, emergency response) at 3 difficulties, each requiring cross-domain coordination. Two instantiations share the interface stack: a **high-fidelity continuous-dynamics suite** (14 gate-admitted scenarios, deception by construction; Section 6.7) and a **semantically enhanced grid-world** (50% feints indistinguishable by numeric features, intercepting a feint is an own-goal; App. H).

4.3 Layered Evaluation Metrics

From ~ 40 raw per-episode measures we compute planning-attributable, execution-attributable, and composite scores (0.5/0.5 default), aggregated with paired statistics (Wilcoxon, sign tests, bootstrap CIs; full definitions: App. E). All stacks run under a *unified no-intelligence regime*—identical briefings, prompts, and channels—so contrasts isolate architecture, not prompt engineering. These scores decompose where performance is earned statically; counterfactual attribution (Section 5.1) is a separate instrument replacing the upper layer to measure causal contribution.

5 Diagnostic Instrumentation

Ranking cannot answer Q3. Unlike the 2×2 deployment accounting identity (Section 3.3), which accounts components statically, counterfactual attribution *replaces* components to measure causal contribution. Two causal tools exploit the HSemi-MDP structure and directly validate the law’s conditions: the goal-dose ablation operationalizes B_{if} causally (condition ii), and the $\Delta P/\Delta E/\Delta I$

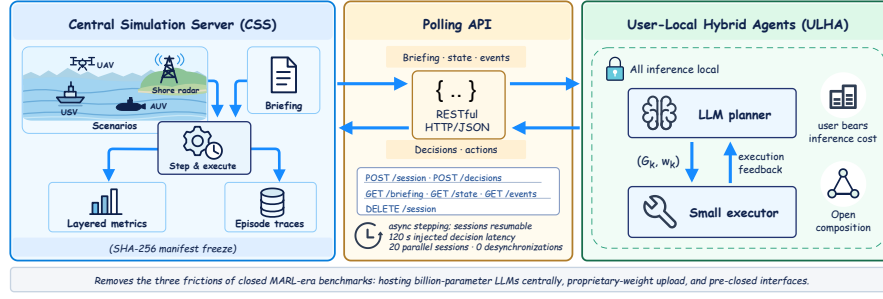


Figure 2: Distributed open evaluation architecture: simulation and metrics central (CSS); all inference local behind a polling API (ULHA).

decomposition—if gains concentrate in ΔP , the LLM’s contribution is in planning (condition i’s mechanism), not execution or interface artifact.

5.1 Counterfactual-Replay Attribution

Method. For a system (π_d, π_e) with performance $V(\pi_d, \pi_e)$, we construct three counterfactuals: (1) *reference-execution replay*—enforce the recorded upper decisions $D = \{(t_m, a_d^m)\}$ at the same times while replacing execution with an independently validated reference executor π_e^* ; (2) *reference planning*—replace π_d with a reference planner π_d^* keeping π_e ; (3) *reference full stack*. In the grid experiments π_d^* is the rule planner, π_e^* the GOAI heuristic executor, and the reference full stack the Rule baseline (App. I). The asymmetry follows from time-scale separation: upper decisions are sparse, so fixing their sequence does not over-constrain; lower actions are dense, so fixing them would erase the policy. The decomposition reads:

$$\Delta E = V(\pi_d, \pi_e) - V(\pi_d, \pi_e^*) \quad (\text{execution}), \quad (2)$$

$$\Delta P = V(\pi_d, \pi_e) - V(\pi_d^*, \pi_e) \quad (\text{planning}), \quad (3)$$

$$\Delta I = V(\pi_d^*, \pi_e^*) - V(\pi_d^*, \pi_e) - V(\pi_d, \pi_e^*) + V(\pi_d, \pi_e) \quad (\text{synergy}). \quad (4)$$

The method is *replay-gated*: a contrast is admitted only after the original episode and every counterfactual pass exact-replay verification—the replayed sequence reproduces the recorded trajectory—so no contrast rests on behavior the system never produced. Positive ΔP (ΔE) means the deployed planner (executor) beats the reference on the same partner; ΔI is the interaction term, maximal when both layers are individually inert but jointly improving.

Validity and preconditions. Five pre-registered checks guard the decomposition (App. D); injected-bottleneck validation localizes 20/20 anchored faults under those gates (Section 6.3). It requires a goal-responsive executor and deterministic dynamics (satisfied by the grid); the real-stream counterfactual validates on live LLM planning streams, while the nondeterministic endpoint uses the prompt-level audit.

5.2 Goal-Dose Interface Ablation

B_{if} is operationalized causally via paired strong-vs-hold interface conditions (mask1/mask2 dose tiers replace goals with legal holds); the strong-hold contrast estimates the goal stream’s total contribution. Replaced commands are legal actions, so the intervention changes what the interface carries, not whether the executor can act. Doses are unit-availability, not bits (App. D).

6 Validation: Boundary Regime and Positive Region

We now validate the entire pipeline and pin down the layering law’s boundary regime. *All numbers derive from released per-episode traces; every table is paired by seed.*

6.1 Setup

Systems: pure RL (MAPPO, goal-conditioned), rule (scripted planner), pure LLM (low-level actions), hybrid (LLM planner over MAPPO or heuristic executors; Qwen3.8-27B, temp 0.1 for grid experiments; high-fidelity evaluation uses temp 0.0). All contrasts are within-batch, paired on shared seeds, with seed-bootstrap 95% CIs (App. F): ten paired seeds per medium-tier batch (goal-dose, six-arm, real-stream), five per complex batch (Section 6.2).

6.2 Grid: Boundary Confirmation

Difficulty cliff. Across tiers, pure MAPPO scores 100% / $\approx 97\%$ / 0% vs. rule 100% / 97% / 65%: the complex tier’s deception load (indistinguishable feints, anomaly relabeling, interference) collapses end-to-end small models yet survives orchestration—exactly the difficulty a deliberator is meant to carry.

The 2×2 factorial isolates the executor floor. On complex, every MAPPO-executor stack scores 0%; with the heuristic executor, rule reaches 65% and LLM 5%. The MAPPO executor is the observed bottleneck under the frozen training protocol; a diagnostic re-training audit (12 configurations) confirms the 0% is robust across current hyperparameters—frozen-medium transfer collapses after on-complex fine-tuning, evidence of training dynamics, not task unsolvability (App. F).

Complex-layer layered stacks confirm the P1 prediction directly. A frozen five-seed paired batch (63101–63105) ran all four

stacks on complex with one fixed medium checkpoint (App. F): LLM+heuristic scores 0.400 vs. Rule 0.787 (paired $\Delta V = -0.387$, CI $[-0.707, -0.053]$), and LLM+MAPPO 0.220 vs. Pure MAPPO 0.240 ($\Delta V = -0.020$, CI $[-0.073, +0.027]$). Two mechanisms surface: where the decision form is a mismatch (geometric micro-control), the LLM planner *hurts*—the same heuristic executor drops from 0.787 (rule goals) to 0.400 (LLM goals); where the executor is the bottleneck (MAPPO at 0%), adding the LLM changes nothing. The medium-tier six-arm accounting shows the same direction ($\Delta V_A = -0.213$), the monotonic worsening being exactly the P1 gradient (App. F).

Tournament. Round-robin Elo: pure RL $\approx$ hybrid (1555 vs. 1553) $>$ rule $\gg$ pure LLM $>$ random—the grid’s boundary outcome, violating DF (geometric interception) and D2 (saturated baseline; App. F).

6.3 Attribution Validation

On injected-bottleneck calibration the method passes all five pre-registered gates (App. D) and localizes 20/20 anchored faults (full table: App. I). A 12-episode natural-failure pilot agrees with human annotation in 4/12 ($\kappa = 0.25$); disagreements are semantic—the ΔI marginal-interaction label is read by humans as “balanced”—confirming ΔI measures layer interaction, not first-order fault location.

6.4 Grid Boundary: Interface Necessary but Insufficient

Three paired batches on medium tier (10 seeds each): goal-dose, six-arm (Table 2), real-stream counterfactual.

The goal interface has a causal contribution under the paired intervention. Disabling it collapses performance: strong—hold = $+0.457$ (rule) / $+0.537$ (LLM), both CIs above 0; under hold both score 0.213 and become indistinguishable; the real-stream counterfactual agrees (10/10 seeds, $+0.420$).

Headroom boundary (P2): interface necessity does not clear a saturated baseline. Against the strongest pure baseline ($V_m = 0.913$), both deployed hybrids sit significantly below it (Table 2)—a near-saturated baseline leaves little headroom. The same prediction holds inside the positive region: IE-03, the only admitted scenario with a near-saturated RL baseline (0.96), is where layering does not pay; across the family layered gain trends negatively with headroom (Spearman $\rho = -0.50$ LLM+Rule, -0.68 LLM+RL; App. A).

The dose–gain curve anchors the law’s increasing branch. Across four paired doses (hold, mask2, mask1, strong), both stacks improve significantly from hold to mask1 but not monotonically beyond—“more interface is always better” is refuted (full table: App. I).

The law’s interface conditions reproduce across models. A four-model scan (two dense, two MoE) reproduces interface necessity in 4/4 and the deployment deficit in 3/4; the fastest MoE reaches the pure-RL level in strong condition—a headroom boundary, not a violation (App. F).

Table 2: Six-arm accounting batch (medium tier, 10 paired seeds): mean composites and per-seed successes, with deployment accounting identities (seed-bootstrap 95% CIs). V_m = per-seed max of pure baselines (rule+rule overall mean 0.913; pure RL overall mean 0.883).

Stack	Mean V	Succ.	Identity	[95% CI]
rule+rule ($V_{RR} = V_m$)	0.913	9/10	—	—
pure RL	0.883	9/10	—	—
rule+RL (V_{RM})	0.810	8/10	$E = -0.103$	$[-0.263, +0.000]$
LLM+rule (V_{LR})	0.700	6/10	$P = -0.213$	$[-0.497, +0.080]$
LLM+RL (V_{LM})	0.673	6/10	$I = +0.077$	$[-0.240, +0.380]$
pure LLM	0.280	1/10	—	—
$B = V_m - V_{RR} = 0$; $\Delta V_A = P - B = -0.213$ $[-0.497, -0.037]$; $\Delta V_B = P + E + I - B = -0.240$ $[-0.490, -0.087]$				

6.5 Decision-Form Anatomy (P1)

Three pre-registered rescue channels—information injection, replanning frequency, interface modality—all fail on the complex tier: the LLM reads and cites the intelligence yet performance does not move, tracing the deficit to semantic-to-geometric conversion under spatial-combinatorial load. The pattern is robust across training configurations (Section 6.2), motivating DF as the law’s first condition. The two negative conditions separate cleanly: interface strength cannot rescue a decision-form mismatch (P1), while a strong interface at a saturated baseline still cannot pay (the fastest MoE reaches only the pure-RL level under strong; Section 6.4, P2)—each prediction bites independently (details: App. H).

6.6 Interface Modality

Interface choice is load-matched (App. H): JSON wins at low semantic load, NL recovers at high load, neither rescues an over-matched planner. A replanning-frequency sweep plateaus at 5-step cycles while cost grows 2.5 $\times$; no frequency reaches the pure-RL baseline—bandwidth cannot substitute for missing headroom (P2).

6.7 From Boundary Conditions to the Gates

The grid results convert into pre-registered acceptance gates (DF/D2/D3; Section 3.3): decision-form match, headroom, and interface budget, with IA measured but not gated. The main run was preceded by determinism gates, channel-fairness audits, and manifest-hash freezing.

Scope and honesty of the gated design. The gates are a *pre-registered design choice*, not post-hoc filtering: they were frozen before the high-fidelity run, so the 14 admitted scenarios are not cherry-picked for layered performance. We acknowledge the circularity: the positive region is confirmed on scenarios selected to satisfy the law’s conditions—the intended test of P3, not an unbiased survey of the family. A 7-scenario sensitivity check bounds the design effect: layered gain is uncorrelated with the DF prior ($\rho = -0.009$, range $[-0.030, +0.086]$), so the law’s predictions do not reduce to the gate threshold itself (App. A). D2/D3-excluded scenarios were not run with layered stacks (that family ships only script policies, and existing weights would conflate “executor

does not know this task” with “no headroom”); the negative side of P3 is thus supported by the grid boundary regime and IE-03 (future work: executor re-training). A headroom probe confirms the family cannot resolve this dimension: in IE-05, degrading planning information collapses every tier to a structural floor ($V \approx 0.18$); P3’s negative side is unstable within-family (App. F).

6.8 Tests 2 and 3: the Positive Region and its Refinement

14 scenarios, five stacks (two layered, end-to-end LLM, two monolithic baselines), unified no-intelligence regime, five seeds per LLM stack, three-part stability criterion (Table 3; Figure 3: end-to-end gap).

Layering wins where the gates hold. Each layered stack beats the Rule baseline in 11 of 14 scenarios and the RL baseline in 12 of 14 (overall 0.77–0.78 vs. 0.63–0.72; Table 3); the stricter every-seed criterion is satisfied in 8/14 LLM+Rule and 9/14 LLM+RL comparisons (17/28 overall). Exceptions: pure RL dominates IE-03 (0.96, minimal headroom); rule wins 3 scenarios per stack. Hybrid advantage is modal, not universal.

Headroom feasibility and conditional-gain analysis. A calibration campaign (960 episodes) confirmed that success-rate saturation does not eliminate score headroom: on IE-09 all pure stacks saturate at 100% SR, yet Rule score drops -0.153 (95% CI $[-0.236, -0.065]$) between quantities 18 and 6. A frozen four-condition confirmation (120 paired episodes, quantities 17/18/19/27) shows LLM+Rule gains $+0.123$ (95% CI $[+0.063, +0.191]$, robust to adjustment) at quantity 17 and $+0.119$ (95% CI $[+0.003, +0.245]$, nominal) at quantity 27, while LLM+RL shows none—hybrid benefit is condition- and executor-dependent. Headroom direction is a diagnostic, not a controlled manipulation (all CIs straddle zero); the claim is delimited in Section 7.2 (protocol and data: supplementary Appendix, Headroom Calibration section).

End-to-end LLM control trails decisively. Pure LLM averages 0.43—half the layered stacks—and trails LLM+Rule on 14/14 scenarios (median deficit 0.33; Figure 3). The same LLM that plans winning allocations cannot convert them into control—consistent with a placement bottleneck rather than a lack of model-level planning capability.

Stripping opponent intelligence does not hurt layered stacks. Removing opponent intelligence (vs. the legacy all-knowing prompt) leaves layered stacks essentially unchanged ($+0.000$ LLM+RL, $+0.021$ LLM+Rule) while end-to-end LLM drops -0.085 ; a prompt audit isolates misleading directives in the briefing channel (supplementary Appendix, Headroom Calibration section). Deliberation-layer value is governed by information *quality*, not *quantity*.

6.9 Mechanism Probe: Layered Value Does Not Require an LLM Information Advantage

We test the competing hypothesis that layered value requires an LLM information advantage with an offline feint-vs-real discriminator probe on the two decoy scenarios (IE-06/IE-11, 400 events each). The result is counterintuitive: the numerical discriminator reaches 84.25%/90.50% balanced accuracy vs. 51.25%/45.00% for the

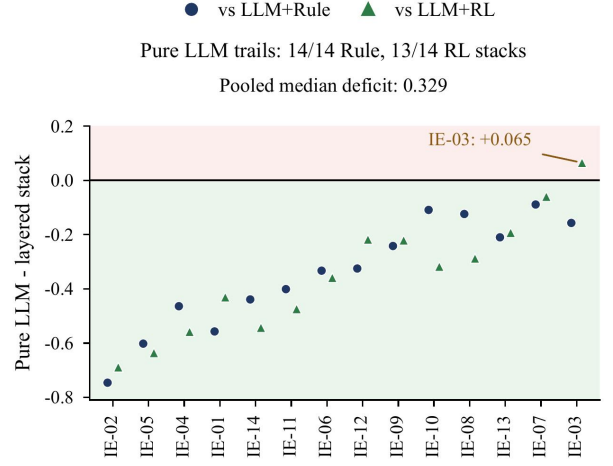


Figure 3: End-to-end LLM control deficit: per-scenario composite-score gap of Pure LLM against each layered stack (frozen gap means, ordered by the average of the two gaps; negative = layered stack ahead). Pure LLM trails LLM+Rule on 14/14 and LLM+RL on 13/14 scenarios, with IE-03 the exception; pooled median deficit 0.33.

offline LLM ($C_{\text{info}} = -0.330 / -0.455$, CIs exclude 0)—the LLM is worse at sensing, yet layered stacks still win on the same scenarios (LLM+Rule $+0.110 / +0.066$ and LLM+RL $+0.136 / +0.138$ vs. the strongest pure baseline). The evidence is thus more consistent with semantic goal decomposition than with an LLM information advantage—why C_{info} is a diagnostic, not a gate. The probe is offline and scenario-local.

7 Discussion

7.1 Insights

I1 — Layering is conditional, and the conditions are measurable. The grid sits at the boundary (decision-form match fails, headroom insufficient); the high-fidelity suite provides evidence for the positive-region hypothesis under the pre-registered design-side admission protocol—falsifiably and by design.

I2 — Executor floor and decision-form mismatch jointly bound layered value. One weak executor floors every planner above it, while the same planners separate under a heuristic executor; on geometrically shaped tasks all three rescue channels fail because the LLM cannot convert semantic understanding into spatial-combinatorial scheduling. Environments crediting LLM deliberation must place difficulty in LLM-strong forms.

I3 — Layered value comes from semantic goal decomposition, not information advantage. The LLM discriminator underperforms a rule baseline ($C_{\text{info}} < 0$), yet layered stacks still win: the evidence is more consistent with the LLM restructuring the task into executable subgoals than with an information advantage.

I4 — Methodological lessons. Pre-registered hypotheses, paired statistics, released traces, and manipulation-validity checks made

Table 3: High-fidelity suite (14 scenarios, unified no-intelligence regime): composite defender score; bold = best. Five seeds per LLM stack (7/11/13/17/19), one replication under nondeterminism; rule/RL never read the briefing (frozen archive, SHA-256 verified); LLM+RL uses one locked checkpoint. Overall: mean $\pm$ sd over 14 scenario means.

Scenario	LLM+Rule	LLM+RL	Pure LLM	Rule	RL
IE-01 single target	0.900	0.774	0.343	0.691	0.876
IE-02 dual threat	0.908	0.851	0.162	0.765	0.757
IE-03 surface raid	0.856	0.634	0.699	0.748	0.960
IE-04 combined arms	0.710	0.804	0.246	0.812	0.687
IE-05 multi-axis	0.790	0.824	0.188	0.763	0.703
IE-06 decoy mixed	0.767	0.793	0.434	0.657	0.652
IE-07 cross-domain	0.792	0.763	0.703	0.749	0.684
IE-08 island strike	0.404	0.568	0.280	0.432	0.301
IE-09 staggered waves	0.880	0.859	0.638	0.820	0.629
IE-10 dual-axis pincer	0.712	0.921	0.603	0.801	0.729
IE-11 decoy screen	0.747	0.820	0.346	0.681	0.184
IE-12 fog onset	0.766	0.659	0.441	0.711	0.641
IE-13 deep strike	0.807	0.790	0.597	0.709	0.538
IE-14 saturation	0.793	0.897	0.354	0.787	0.506
Overall	0.774 $\pm$ 0.124	0.783 $\pm$ 0.100	0.431 $\pm$ 0.187	0.723 $\pm$ 0.098	0.632 $\pm$ 0.203
Mean wins vs. Rule	11	11	0	–	–
Mean wins vs. RL	12	12	4	–	–

the boundary confirmation productive: the grid seeded the gates that let the high-fidelity suite (admitted by pre-registered design-side gates) confirm the positive region. Briefing channels should be audited as interface design—the no-intelligence audit isolated misleading directives.

7.2 Limitations and Threats to Validity

Scope of the framework. The layering law is domain-agnostic: it applies to any problem whose structure separates a slow, semantic deliberation layer from a fast, continuous execution layer joined by a goal interface; problems without such a time-scale separation are outside its scope by construction. Our instantiations validate both sides (boundary regime and positive region), where the three conditions double as an operational scope test.

Scope and power. The grid is minimal-geometry (boundary regime only); high-fidelity covers 14 of 36 interception scenarios, five stacks, one prompt regime. Goal doses are unit-availability, not bits; we do not fit f in Eq. 1. Five seeds per LLM stack, three-part stability as the primary gate, Wilcoxon tests corroborating key contrasts ($p < 0.002$).

Trust and instrumentation. Distributed evaluation cannot verify claimed architectures; rate limits bound but do not eliminate exploitation. Attribution is calibrated on injected and anchored faults; blind location of natural faults is a disclosed boundary (App. I). Replay gates guarantee a recorded trace reproduces under the same configuration, not LLM bit-identical re-emission; the nondeterministic endpoint uses the prompt-level audit and stability criterion (App. D).

Decision-form and headroom delimitation. Decision-form match is a design-side gate anchored by the grid/high-fidelity contrast and the mechanism probe; a 7-scenario check of the DF prior ($\rho = -0.009$) is too low-powered to claim predictive validity (mid-range validation: App. A). Headroom is similarly delimited: E1 (120 paired episodes) supports LLM+Rule gains at two of four frozen quantities, but separated tiers were not achieved (95%

CI's straddle zero): calibration feasibility and partial direction confirmation, not a causal test, supported by the boundary regime (P2) and the gain–baseline trend (App. A).

8 Conclusion

We formalize decision-stack bifurcation (HSemi-MDP), derive measurable coupling, and distill the conditional layering law: decision-form match, interface carrying, baseline headroom. Validation is two-sided: the grid confirms the boundary prediction (no pay when decision-form match fails and headroom is insufficient), the gate-admitted suite (admitted by pre-registered design-side gates DF/D2/D3, not post-hoc endpoint re-certification) provides evidence for the positive-region hypothesis, and a mechanism probe locates layered value in semantic goal decomposition, not an LLM information advantage. In the evaluated cross-domain setting, the evidence supports placing the LLM at the planning layer above a mature executor rather than using end-to-end LLM control, when the three preconditions jointly provide room for layered gains; environments, traces, and code are released.

References

- [1] Norah A. Alzahrani, Hisham Abdullah Alyahya, Yazeed Alnumay, Sultan Al-rashed, Shaykhah Z. Alsubaie, Yousef Almushayqih, Faisal Abdulrahman Mirza, Nouf M. Alotaibi, Nora Altwairesh, Areeb Alowisheq, M. Saiful Bari, and Haidar Khan. 2024. When Benchmarks are Targets: Revealing the Sensitivity of Large Language Model Leaderboards. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- [2] Anton Bakhtin, Noam Brown, Emily Dinan, Gabriele Farina, Colin Flaherty, Daniel Fried, Andrew Goff, Jonathan Gray, Hengyuan Hu, Athul Paul Jacob, Mojtaba Komeili, Karthik Konath, Minae Kwon, Adam Lerer, Mike Lewis, Alexander H. Miller, Sasha Mitts, Adithya Renduchintala, Stephen Roller, et al. 2022. Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning. *Science* 378, 6624 (2022), 1067–1074.
- [3] Xiaohe Bo, Zeyu Zhang, Quanyu Dai, Xueyang Feng, Lei Wang, Rui Li, Xu Chen, and Jirong Wen. 2024. Reflective Multi-Agent Collaboration based on Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [4] Marc Bou, Pierluca D’Oro, Soumya Kar, et al. 2023. BenchMARL: A Benchmark for Multi-Agent Reinforcement Learning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- [5] Junzhe Chen, Xuming Hu, Shuodi Liu, Shiyu Huang, Wei-Wei Tu, Zhaofeng He, and Lijie Wen. 2024. LLMARENA: Evaluating Large Language Models in Dynamic Multi-Agent Environments. In *Proceedings of the 62nd Annual Meeting of*

- the Association for Computational Linguistics (ACL).
- [6] Charles Dansereau, Junior-Samuel Lopez-Yepe, Karthik Soma, and Antoine Fagette. 2025. The Heterogeneous Multi-Agent Challenge. In *Proceedings of the 28th European Conference on Artificial Intelligence (ECAI)*.
 - [7] Benjamin Ellis, Mikayel Samvelyan, Robert Faulkner, et al. 2023. SMACv2: An Improved Benchmark for Cooperative Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
 - [8] Lasse Espeholt, Hubert Soyer, Rémi Munos, et al. 2018. IMPALA: Scalable Distributed Deep-RL with Importance Weighted Actor-Learner Architectures. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*.
 - [9] Jakob N. Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. 2018. Counterfactual Multi-Agent Policy Gradients. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.
 - [10] Qingxu Fu, Zhiqiang Pu, Min Chen, Tenghai Qiu, and Jianqiang Yi. 2024. Prioritized League Reinforcement Learning for Large-Scale Heterogeneous Multiagent Systems. *arXiv preprint arXiv:2403.18057* (2024).
 - [11] Minghong Geng, Shubham Pateria, Budhitama Subagdjia, Lin Li, Xin Zhao, and Ah-Hwee Tan. 2025. L2M2: A Hierarchical Framework Integrating Large Language Model and Multi-agent Reinforcement Learning. In *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*.
 - [12] Ran Gong, Qiuyuan Huang, Xiaojian Ma, Yusuke Noda, Zane Durante, Zilong Zheng, Demetri Terzopoulos, Fei-Fei Li, Jianfeng Gao, and Hoi Vo. 2024. MindAgent: Emergent Gaming Interaction. In *Findings of the Association for Computational Linguistics: NAACL 2024*.
 - [13] Siyi Hu, Yifan Zhong, Minquan Gao, Weixun Wang, Hao Dong, Xiaodan Liang, Zhihui Li, Xiaojun Chang, and Yaodong Yang. 2023. MARLlib: A Scalable and Efficient Multi-Agent Reinforcement Learning Library. *Journal of Machine Learning Research* 24, 315 (2023), 1–23.
 - [14] Ziqi Jia, Junjie Li, Xiaoyang Qu, and Jianzong Wang. 2025. Enhancing Multi-Agent Systems via Reinforcement Learning with LLM-based Planner and Graph-based Policy. *arXiv preprint arXiv:2503.10049* (2025).
 - [15] Subbarao Kambhampati, Karthik Valmeekam, Lin Guan, Mudit Verma, Kaya Stechly, Siddhant Bhambri, Lucas Saldyt, and Anil Murthy. 2024. Position: LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.
 - [16] Karol Kurach, Anton Raichuk, Piotr Stańczyk, Michał Zajac, Olivier Bachem, Lasse Espeholt, Carlos Riquelme, Damien Vincent, Marcin Michalski, Olivier Bousquet, and Sylvain Gelly. 2020. Google Research Football: A Novel Reinforcement Learning Environment. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*.
 - [17] Joel Z. Leibo, Edgar Dueñez-Guzmán, Alexander Sasha Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Köster, Jayd Matyas, Charles Beattie, Igor Mordatch, and Thore Graepel. 2021. Scalable Evaluation of Multi-Agent Reinforcement Learning with Melting Pot. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research (PMLR), Vol. 139)*.
 - [18] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and Minlie Huang. 2024. AgentBench: Evaluating LLMs as Agents. In *International Conference on Learning Representations (ICLR)*.
 - [19] Andrei Lupu, Timon Willi, and Jakob Foerster. 2025. The Decrypto Benchmark for Multi-Agent Reasoning and Theory of Mind. *arXiv preprint arXiv:2506.20664* (2025).
 - [20] Prasanna Maddila, Eric Casellas, Patrick Chabrier, Régis Sabbadin, and Meritxell Vinyals. 2024. APE: An Anti-Poaching Multi-Agent Reinforcement Learning Benchmark. In *Deep Reinforcement Learning Workshop (EURL)*.
 - [21] Grégoire Mialon, Clémentine Fourrier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. 2024. GAlA: A Benchmark for General AI Assistants. In *International Conference on Learning Representations (ICLR)*.
 - [22] Volodymyr Mnih, Adrià Puigdomènech Badia, Mehdi Mirza, et al. 2016. Asynchronous Methods for Deep Reinforcement Learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*.
 - [23] Mahmoud Mohammadi, Yipeng Li, Jane Lo, and Wendy Yip. 2025. Evaluation and Benchmarking of LLM Agents: A Survey. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.
 - [24] Héctor Muñoz Avila, David W. Aha, and Paola Rizzo. 2025. ChatHTN: Interleaving Approximate (LLM) and Symbolic HTN Planning. In *Proceedings of the International Conference on Neuro-symbolic Systems (NeuS) (Proceedings of Machine Learning Research (PMLR), Vol. 288)*.
 - [25] Arun Nair, Praveen Srinivasan, Sam Blackwell, et al. 2015. Massively Parallel Methods for Deep Reinforcement Learning. *arXiv preprint arXiv:1507.04296* (2015).
 - [26] Maria Rigaki, Ondřej Lukoš, Carlos A. Catania, and Sebastian Garcia. 2023. Out of the Cage: How Stochastic Parrots Win in Cyber Security Environments. *arXiv preprint arXiv:2308.12086* (2023).
 - [27] Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Nantas Nardelli, Tim G. J. Rudner, Chia-Man Hung, Philip H. S. Torr, Jakob Foerster, and Shimon Whiteson. 2019. The StarCraft Multi-Agent Challenge. In *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*.
 - [28] Risal Shahriar Shefin, Debashis Gupta, Thai Le, and Sarra Alqahtani. 2026. Interpretable Failure Analysis in Multi-Agent Reinforcement Learning Systems. *arXiv preprint arXiv:2602.08104* (2026).
 - [29] Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. 2023. HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face. In *Advances in Neural Information Processing Systems (NeurIPS)*.
 - [30] Philipp Dominik Siedler. 2025. HIVEX: A High-Impact Environment Suite for Multi-Agent Research. *arXiv preprint arXiv:2501.04180* (2025).
 - [31] Theodore Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2024. Cognitive Architectures for Language Agents. *Transactions on Machine Learning Research (TMLR)* (2024).
 - [32] Richard S. Sutton, Doina Precup, and Satinder Singh. 1999. Between MDPs and Semi-MDPs: A Framework for Temporal Abstraction in Reinforcement Learning. *Artificial Intelligence* 112, 1–2 (1999), 181–211.
 - [33] Justin Terry, Benjamin Black, Nathaniel Grammel, et al. 2021. PettingZoo: Gym for Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
 - [34] Stelios Triantafyllou, Aleksa Sukovic, Yasaman Zolfimoselo, and Goran Radanovic. 2025. Counterfactual Effect Decomposition in Multi-Agent Sequential Decision Making. In *Proceedings of the 42nd International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research (PMLR), Vol. 267)*. 60072–60098.
 - [35] Karthik Valmeekam, Matthew Marquez, Sarath Sreedharan, and Subbarao Kambhampati. 2023. On the Planning Abilities of Large Language Models: A Critical Investigation. In *Advances in Neural Information Processing Systems (NeurIPS)*.
 - [36] Erich C. Walter, Kimberly J. Ferguson-Walter, and Ahmad D. Ridley. 2021. Incorporating Deception into CyberBattleSim for Autonomous Defense. *arXiv preprint arXiv:2108.13980* (2021).
 - [37] Xiaofei Wang, Zehui Xiong, Chenhan Xu, et al. 2023. Federated Reinforcement Learning: A Survey. *IEEE Transactions on Neural Networks and Learning Systems* (2023).
 - [38] Abraham Itzhak Weinberg. 2025. MACIE: Multi-Agent Causal Intelligence Explainer for Collective Behavior Understanding. *arXiv preprint arXiv:2511.15716* (2025).
 - [39] Yingxuan Yang, Bo Huang, Siyuan Qi, Chao Feng, Haoyi Hu, Yuxuan Zhu, Jinbo Hu, Haoran Zhao, Ziyi He, Xiao Liu, Zongyu Wang, Muning Wen, Lin Qiu, Xuezhi Cao, Xunliang Cai, Yong Yu, and Weinan Zhang. 2026. Attribution-Based Analysis and Optimization of Modular Agentic Workflows. In *Findings of the Association for Computational Linguistics: ACL 2026*.
 - [40] Jianzhu Yao, Kevin Wang, Ryan Hsieh, Haisu Zhou, Tianqing Zou, Zerui Cheng, Zhangyang Wang, and Pramod Viswanath. 2025. SPIN-Bench: How Well Do LLMs Plan Strategically and Reason Socially? *Proceedings of the Conference on Language Modeling (COLM)* (2025).
 - [41] Meng Yao, Qiyue Yin, Jun Yang, Tongtong Yu, Shengqi Shen, Junge Zhang, Bin Liang, and Kaiqi Huang. 2021. The Partially Observable Asynchronous Multi-Agent Cooperation Challenge. *arXiv preprint arXiv:2112.03809* (2021).
 - [42] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR)*.
 - [43] Rui Ye, Wenhao Wang, Jingyi Chai, Dihan Li, Zexi Li, Yinda Xu, Yaxin Du, Yanfeng Wang, and Siheng Chen. 2024. OpenFedLLM: Training Large Language Models on Decentralized Private Data via Federated Learning. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 6137–6147.
 - [44] Chiyu Zhang, Marc-Alexandre Côté, Michael Albadawy, Anush Sankaran, Jack W. Stokes, Tong Wang, Amir Abdi, William Blum, and Muhammad Abdul-Mageed. 2025. DefenderBench: A Toolkit for Evaluating Language Agents in Cybersecurity Environments. *arXiv preprint arXiv:2506.00739* (2025).
 - [45] Shuyang Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. WebArena: A Realistic Web Environment for Building Autonomous Agents. In *International Conference on Learning Representations (ICLR)*.
 - [46] Guobin Zhu, Rui Zhou, Wenkang Ji, and Shiyu Zhao. 2025. LAMARL: LLM-Aided Multi-Agent Reinforcement Learning for Cooperative Policy Generation. *arXiv preprint arXiv:2506.01538* (2025).
 - [47] Kunlun Zhu, Hongyi Du, Zhaochen Hong, Xiaocheng Yang, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Xiangru Tang, Heng Ji, and Jiaxuan You. 2025. MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.